

Slide 4 The enzyme active area is like a magic pocket, after putting the reactant in it, it can become the product. Of course we already know that this pocket can lower the activation energy of the reaction intermediate, but how does it do it? This illustration suggests four possible mechanisms.

(1) In the structure of the transition state molecules, there are often quite high local charges, and the appropriate positions in the catalytic pocket are just arranged with groups that can neutralize the local high charges, thus stabilizing the transition state.

(2) In aqueous solution, the bonding or reaction between many ions will be disturbed by water molecules, so the concave catalytic pocket can isolate most of the water molecules, making the reaction between ions go smoothly.

(3) Some of the amino acid groups in the active region can be due to the special spatial arrangement, so that the originally low-reactivity group (such as Ser-OH) can be taken away due to the influence of other nearby groups (such as His Its H⁺), and become a highly reactive group (such as Ser-O⁻).

(4) The active region is usually also the binding region of the accessory substance, and the accessory molecule has a strong charge group, which can directly participate in the reaction or assist the reaction.

Slide 6 The imaginary enzyme of Stickase in Lehninger's textbook is an excellent example, please think about it more.

Note that the shape of the stickase-matrix binding region is not directly complementary to the matrix, but has a better conformational complementarity to its "intermediate transition" state. In addition, the binding area between the

enzyme and the substrate is mostly deeply embedded in the molecule, which is a major advantage and feature of the enzyme action.

Slide 7 Generally, the main function of a catalyst is to provide a surface space for reactants to react on, and this surface can provide a space convenience, making it easier for reactants to transform into products. In the process of converting reactants into products, there will be an intermediate transition state between the two, which is the most difficult step in the entire catalytic reaction. Usually, this catalytic space can make this transition state more stable and easier form. The same is true for enzymes. The active area of enzymes provides a catalytic pocket, which can stabilize the formation of the intermediate transition state; that is, enzymes can make the transition state easier to form, and the required energy (activation energy) is relatively low.

Note, however, that having a catalytic surface alone is not enough, as the example of catalytic antibodies (abzymes) illustrates.

Slide 15 An important reason why the active region is buried deep inside is that the catalytic reaction must avoid water molecules, lest the reaction be disturbed by hydration, proper bonding and proton or electron transfer (water molecules are too powerful). An important reason why the active region is buried deep inside is that the catalytic reaction must avoid water molecules, lest the reaction be disturbed by hydration, proper bonding and proton or electron transfer (water molecules are too powerful).

Slide 22 A carbon atom has four outer electrons, so the sp^3 orbital can be bonded to four atoms. For example, the carbon atom in the sp^3 orbital in the center of the left figure above can be bonded to A, B, C, and D atoms. This electron orbital contributes to at least two facts: (1) a carbon atom can bond to four other atoms, the most common of which is to another carbon, which is fundamental to organic chemistry; (2) a carbon atom not only The number of bonded atoms is large, and when the surrounding atoms or groups are different, the central carbon atom will become an asymmetric carbon, and stereoisomers will be produced. Stereoisomers are two or more molecules that have the same molecular formula but different three-dimensional arrangements. As shown in the above right picture, from the above A, although they are all B, C, D, there may be two different arrangements (clockwise or counterclockwise). The three-dimensional

space of these two molecules. The arrangement is different, the two cannot overlap together. These two stereoisomers are in fact mirror images of each other and are specifically called enantiomers.

The most common molecule with stereoisomers is amino acid, the α carbon on the molecule is an asymmetric carbon; therefore, except glycine, all amino acids can have D-, L- isomerism. Most of the substances in nature only use L-amino acids. Enzymes also recognize the difference between the two molecules, and most use only one of them.

Slide 26 Model of the catalytic domain and the specificity recognition domain in the active domain, please note that the identified amino acid group is the group (R) on the N-terminal side of the peptide bond. The specific recognition region of Chymotrypsin forms a fairly large pocket, and there are no polar amino acids in the pocket, so it prefers the position of hydrolyzing Trp, Phe, Tyr.

Slide 27 The Ser protein family uses its specificity region to recognize the type of substrate, and the characteristics of the substrate determine the configuration of the recognition region. Trypsin has a slender specificity pocket and an Asp 189 at the bottom, so its substrates are all positively charged elongated amino acids such as Lys and Arg. Conformation does determine the function and role of many proteins.

Slide 59 The figure above sorts out the gene expression process in prokaryotic cells and eukaryotic cells, and points out the control points that can be regulated in the whole process. Many of them are regulated at the DNA or RNA level. Cells can turn on or off the expression of the corresponding protein by switching a certain gene. The control method of this kind of gene expression is very complicated; most of them use some kind of regulatory protein to direct whether the gene can be transcribed into mRNA at the front end of the gene (regulatory region). Therefore, the regulation of enzyme activity in cells has many different levels. The focus of this course is on the modification and control of proteins after protein synthesis. As for the regulation mechanism of gene expression level, it is the main axis of molecular biology. National Taiwan University There are a number of related courses available as electives.

Slide 60 The above figure sorts out the regulation methods of the five enzymes, among which the method and mechanism of using inhibitors to inhibit the

enzymes have been introduced in the fourth section and will not be discussed again. The remaining four are all reversible adjustments except for bond cleavage (6.1), which is an irreversible modification method. Among the three reversible regulation methods, only phosphorylation (6.2) is a covalent modification, and the other two are non-covalent combinations, which are modified by combining certain molecules with enzymes; Among them, cAMP and calmodulin (6.3) are both information transduction molecules, which are the intermediaries that transmit instructions from the outside of the cell to the inside; the other feedback control (6.4) is to control the activity of enzymes through the upstream and downstream metabolites in the cell. These methods are all working hard to regulate the activity of enzymes inside and outside the cells in the organism at the same time, so that the cells can achieve the most effective and freely controllable physiological functions.

In recent years, great progress has been made in the regulation of the activity of enzymes, especially the way of information transmission is extremely complex, and its variety is even more dazzling. This course is just an introduction, so try to simplify the various topics to be introduced, usually using a more mature example or mechanism as the protagonist of the story to illustrate, and point out the most important main axis of the topic; as for how deep it is, usually Enough is enough, and its depth and breadth depend on the efforts of the students themselves. Recently, National Taiwan University has already offered many related courses to discuss in-depth information transmission. Those who are interested should be able to choose appropriate courses for further research.

Slide 61 Information transduction is the process by which cells transmit external information into cells and amplify it.

It seems that a messenger came from outside the city (for example, some kind of hormone), and at the gate of the city, he gave a message to the person guarding the gate (hormone receptor), and the latter passed the message to the people below the ten gates (probably G protein). Each of

the ten people will pass it on to another ten people (probably cyclase), so that it will spread throughout the entire cell.

Slide 62 Why does phosphorylation of GP, or binding to effector molecules, alter its activity? The main reason is because GP can change its molecular configuration due to modifications such as phosphorylation. The molecular configuration of GP has two forms, one is the relatively loose R form, which has greater activity; the other is the tighter T form, which has lower activity. When GP is phosphorylated or bound to a small molecule, it switches between these two configurations, thereby altering its activity. The protein is phosphorylated by using the -OH group on Ser, Thr or Tyr in its amino acid sequence. After phosphorylation, these amino acid groups become extremely negatively charged, which will cause the attraction between the positively charged groups thus pulls the molecule to change its configuration. It was later found that the imidazole group on His is also phosphorylated.

Phosphorylation and dephosphorylation of proteins are quite common phenomena, but whether the activity increases or decreases after phosphorylation is added varies with different proteins or cells.

Phosphorylation can be said to be a weak method for proteins to supplement their twenty basic amino acid groups, so that proteins have stronger groups that can be used; but even better than this is to use phosphorylation that is modified later, It can also be removed with enzymes, and the result becomes a freely transformable modification

method that can flexibly regulate enzyme activity, which can be called "phosphorylation-dephosphorylation cycle" control.

Slide 63 Too high or too low blood sugar will have a bad effect on the human body, so there is an extremely complex blood sugar regulation system in the body. The protagonists are glycogen synthase (GS), which synthesizes glycogen, and degrades glycogen to produce glucose. Glycogen phosphorylase (GP). These two enzymes are respectively affected by the hormones insulin (insulin) and glucagon (glucagon), and the human body can use the control hormone concentration to regulate blood sugar concentration.

When these two hormone molecules reach the target cell, there is a receptor for this hormone on the cell membrane. After the two are combined, the information can be transmitted into the cell, triggering the activity of ferment, and carrying out the required carbohydrate metabolism. How is this information transmitted to the final enzyme GS and GP? This is the recent extremely vigorous information transmission and the typical enzyme regulation mechanism. The following will take glycogen phosphate enzyme as the main object and example to illustrate how the enzyme regulates its activity through phosphorylation, information transduction molecules and feedback control.

Glycogen phospholytic enzymes have the above-mentioned various regulatory mechanisms, and few enzymes are regulated so intensively, because glycogen phospholytic enzymes are so important in the utilization

of carbohydrates. Let's first understand the molecular structure and biochemical properties of glycogen phosphorylase enzymes.

Slide 64 When the glucose concentration in human blood rises, in just a few minutes, the blood concentration of at least two enzymes will change drastically. The concentration of glycogen phosphorylase enzymes drops quickly, while glycogen synthase enzymes rise sharply, all within a few minutes. From this observation, there are two notable inferences or facts.

(1) The glucose concentration in the blood is a signal that can quickly induce certain responses.

(2) The ebb and flow of the above two enzymes within a few minutes is unlikely to be regulated by a gene switch, and it is more likely that the activity is controlled by direct modification of the protein.

The whole thing is caused by the hormonal regulation system described on the previous page, which induces a signaling response in its target cells.

Slide 65 Glycogen phosphorylase (GP) can perform phosphorolysis reaction on glycogen, start from the end of glycogen long chain, cut off glucose one by one, and add phosphoric acid to become Glc-1-P, which can enter glycolysis act to produce energy.

Normally, GP exists in the less active b-type, and it needs to be phosphorylated by its dedicated kinase (GP kinase) to become the active a-type. And this kinase will be active only after being phosphorylated by another kinase (protein kinase A, PKA) to dephosphorylate GP; and PKA will not be active until it is activated by cAMP. cAMP is an important

information transduction molecule. When the hormone glucagon binds to the target cell, it induces the G protein on the inner side of the cell membrane on the receptor of the cell membrane, and the latter activates ATP cyclase to make ATP produce cAMP. Therefore, this path transmits the information from external hormones to PKA via cAMP, and then to GP kinase, then activates GP to start performing cell tasks, breaking down glycogen into simple sugars for use in cells or the human body need.

The above-mentioned phosphorylation is just a regulation method of GP, which is a covalently bonded reversible modification, because the phosphorylated GP can also use phosphatase to remove phosphoric acid to restore its inactive type b. In addition, GP activity is also affected by its metabolites, for example, when there are too many monosaccharides or energy molecules in blood sugar, it does not need to hydrolyze glycogen; this monosaccharide or energy molecules (such as Glc-6-P, ATP) Can directly bind to GP to rapidly reduce its activity.

Therefore, in addition to phosphorylation, the activity of GP can also be regulated by small molecules such as AMP and Glc-6-P; Therefore, it is called ectopic enzyme.

Slide 66 Protein kinase A (PKA) in protein kinase can phosphorylate a wide range of proteins, and its activity is activated by cAMP. In some cells, PKA enters the nucleus and phosphorylates proteins that promote gene transcription (such as the CRE-binding protein above, CREB), turning certain genes on.

PKA has four subunits, including two catalytic subunits (C) and two regulatory subunits (R), which release active catalytic subunits upon receiving cAMP.

Slide 67 Therefore, the regulation of the activity of glycogen phosphorylase enzymes is intertwined into a complex cascading structure (as shown in the figure above). In the cell, cAMP starts to communicate, and finally reaches glycogen phosphorylase enzymes to achieve the goal of glycogen decomposition and utilization. But how is the message coming in from outside the cell? Take the hormone glucagon as an example. When glucagon finds a cell with a suitable receptor on its surface and binds to it, a series of reactions occur on the cell membrane: first, after the receptor receives the hormone, it induces the activation of G protein (the detailed mechanism is in the next figure), G protein then activates adenylate cyclase, which can convert ATP into cAMP, and you probably know the story below.

There is an important point here, which is the receptor of the above cells. Receptor is like the signboard of a store. Those who have the signboard of the coffee shop can drink coffee, and those who have the signboard of beef noodles can eat noodles. The same is true for cells. There are cells with different functions, and the receptors on their surface can inform the correct information of hormones. Therefore, the connection between the two is also very specific, a bit like a key and a door lock. What kind of key opens its corresponding door lock cannot be mistaken. With such a mechanism,

hormones such as glucagon can find the correct object, transmit information to them, and cause the correct response.

Slide 68 How is information transmitted on the cell membrane? There are really many ways to do it. Generally, according to the type of receptor, three main types of modes can be sorted out:

(1) G-protein-linked Receptor: Receptor is activated after linking with G protein. G protein itself is a GTP-binding protein. After binding to GTP, it can activate adenylate cyclase, which catalyzes ATP into cAMP.

(2) Enzyme-linked Receptor: Receptor is linked to activator (such as tyrosine kinase), which activates itself by autophosphorylation, and also attracts some enzymes (such as phosphatase containing SH2 domain) and activates it, and then continues the downstream catalytic pathway .

(3) Ion-channel-linked Receptor: It is related to the active transport of ions on the cell membrane.

Slide 69 The receptor on the cell surface is very important because it marks the activity in the cell to receive the correct external information and make the correct physiological response. Take the cells that metabolize glycogen as an example. Cells that can accept glucagon will trigger a series of reactions in the cells to produce cAMP to activate PKA. This PKA then phosphorylates several enzymes. The result of phosphorylation increases the activity of certain enzymes. (such as GP kinase, protein phosphatase inhibitor), but also reduce the activity of certain enzymes (such as glycogen synthase, protein phosphatase). The final

result is to reduce the synthesis of glycogen and increase the degradation of glycogen so that the body can use glucose to generate energy.

The protein phosphatase in the above figure will remove phosphorylation (see the curved upward dotted arrow in the above figure), thereby reducing the activity of glycogen phospholytic enzymes and increasing the activity of glycogen synthetase, which is opposite to the desired result of glucagon, so in this cell Inhibited (although it is also phosphorylated, but the activity is reduced after phosphorylation). Conversely, the inhibitor of this enzyme is activated, which further controls the activity of protein phosphatase.

Slide 70 Information conduction has a certain conduction route, and which molecule is transmitted to that one, all have their specificity; for example, cyclase specifically catalyzes the production of cAMP in large quantities. This diagram divides this pathway into several levels, and finally reaches the effector cAMP, which will produce a specific effect in the cell to perform the negative physiological function, also known as the second messenger. Although the information conduction in each cell is quite specific, the conduction routes between different cells are very flexible, and the above-mentioned conduction units at various levels can be combined in various ways. That is to say, each layer has many possible choices, and cells can choose a certain conductive substance, and they can have different effects. For example, it is also the phosphorylation of kinases, but some proteins are phosphorylated to increase activity, and some are just the opposite, depending on what kind of protein transduction molecules are in the cell.

Therefore, in the design of information transmission, cells not only take into account their work efficiency and use the amplification effect of enzymes to amplify information (amplification); Combined into a variety of different conduction routes to meet the various needs of cells (flexibility).

Slide 81 In fact, the kinetic experiment is very simple. Different amounts of substrates are added to each test tube, and the same amount of enzyme is used to catalyze the reaction, and the resulting products are measured within a fixed time. It can be found from the results that as the substrate concentration increases, the reaction becomes faster, but when the substrate concentration is too high, the activity of the enzyme (product concentration) cannot rise any further.

The result is directly plotted as the above curve, which is a part of the hyperbola and can be described by mathematical formula.

Slide 82 In fact, students are very much like enzymes. They study hard to get good test results, but each student's reading or test performance is different (such as students A, B and C in the above picture). Generally, the more chapters (matrix concentration) in the exam, the better the efficiency of students preparing for the exam (chapters 1-3), but if the scope of the exam is too much (chapter 4 or more), candidates will usually be unable to cope and can only reach A saturation point (or simply give up).

At the same time, if students are tested with different chapters, different students will have different performances. For example, the performance curves of each student above are different; Higher than Jiasheng. The

same is true for enzymes. Some enzymes perform better, while others are more lazy, and each has a different performance curve.

When observing these situations, one fact can be vaguely discovered, that is, most students will start to pick up their textbooks before the exam, no matter how well they do in the exam. This is also the most basic starting point of the subsequent kinetic research. Enzymes must first be combined with the substrate before they can catalyze.

Slide 94 Please use the above data and make your own graph to see if it matches the result graph.

This experiment should have a group of control group (number 0), without adding substrate ($[S] = 0$), if the absorbance value obtained after the reaction of Fajimase is the background value of this reaction, the above data should have been deducted This background value.

Slide 113 These inhibition mechanisms can be described by enzyme kinetics, and the use of double reciprocal plots can clearly indicate which inhibition mode they belong to. However, the above three kinds of graphs are the most typical ones. In many cases, the graphing results obtained by experiments may have mixed patterns, which are more complicated suppression mechanisms or other interference factors.



